# COSC 39: Theory of Computation

**Problem 1**. **Punched-tape machine**
Punched-tape machine is a Turing machine equipped with an infinite two-way single-row tape, where each cell can be either punched or unpunched. Furthermore, once a cell is punched, it can never be unpunched. Alternatively, a punched-tape machine is just a Turing machine with unary alphabet consisting of a single symbol ■ (together with the empty cell symbol □), where any symbol ■ written on the tape can never be erased. Prove that the punched-tape machine is equivalent in power to the standard Turing machine. In other words, the languages accepted by the punched-tape machines are exactly all the computable languages.

*Solution.*

Let $M$ be a punched-tape machine; our goal is to show that $M$ can simulate a standard Turning machine $M'$ with a two-way infinite tape and the binary alphabet.

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